

Solution: $[Qr_1 \dots Qr_p] = Q[r_1 \dots r_p]$ (By defn. of matrix multiplication). $\because$ each of the vectors $r_1, \dots, r_p$ are vectors in $\mathbb{R}^n$, $\therefore [r_1 \dots r_p]$ is a $n \times p$ matrix. $\square$

Exercises 15 and 16 concern arbitrary matrices A, B and C for which the indicated sums and products are defined. Mark each statement True or False. Justify each answer.

15a) If A and B are 2×2 matrices with columns a_1, a_2 and b_1, b_2 , respectively, then $AB = [a_1 b_1 \ a_2 b_2]$

Solution: By the defn. of matrix multiplication on page 97, $AB = [Ab_1 \ Ab_2]$

By the Defn. of AB on page 35, $AB = [Ab_1 \ Ab_2] \neq [a_1 b_1 \ a_2 b_2]$.
 $\therefore$ The given statement is False.

b) Each column of AB is a linear combination of the columns of B using weights from the corresponding column of A .

Solution: From the bottom most grey box of page 97, Each column of AB is a linear combination of the columns of A using weights from the corresponding column of B .

$\therefore$ The given statement is False.

c) $AB+AC = A(B+C)$

Solution: This statement is True from Theorem 2(b) on page 99.

d) $A^T + B^T = (A+B)^T$

Solution: This statement is True from Theorem 3(b) on page 101

e) The transpose of a product of matrices equals the product of their transposes in the same order.

Solution: From Theorem 3(d) on page 101, we know that the given statement is False.